Linear Equations: Learning with Spaghetti

A hands-on introduction to lines and systems of equations

Learning goal: Understand that a linear equation in two variables describes a straight line, and that solving two linear equations means finding points that satisfy both.

Materials

Two pieces of uncooked spaghetti, graph paper, pencil, and ruler.

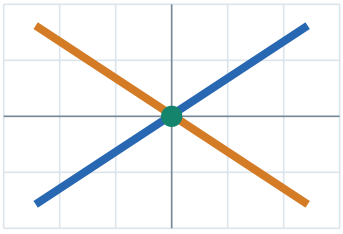
1. Start with one piece of spaghetti

Place one straight piece of spaghetti on graph paper. Imagine extending it forever in both directions. A straight line contains infinitely many points. A linear equation in two variables, such as **y = x + 1**, describes a line: every point on that line makes the equation true.

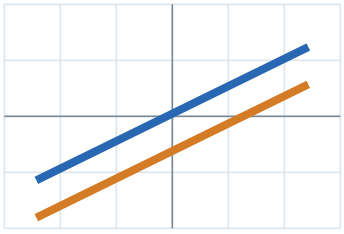
Ask: Can you tilt the spaghetti and still have a straight line? How many points are on a line?

2. Add a second piece

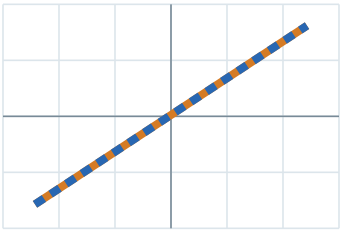
Place a second strand on the paper. Move the strands so they cross, then make them parallel, then place one directly over the other. What changes?



Intersecting: one solution



Parallel: no solution



Same line: infinitely many

Ask: If each strand represents an equation, what would a crossing point mean? Can two distinct straight lines cross twice?

3. Connect the model to a system

A **system of linear equations** is two or more linear equations considered together. A **solution** must make every equation true at the same time. On a graph, it is a point shared by both lines.

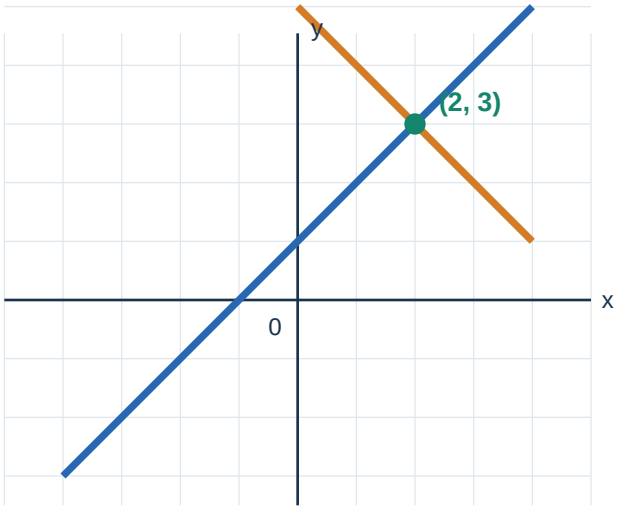
Key idea: Two lines may intersect once, never intersect, or lie exactly on top of each other. Those cases give one, zero, or infinitely many solutions.

From Spaghetti to Equations

Consider the system: **y = x + 1** and **y = -x + 5**. Each equation makes a line. Where do they meet?

y = x + 1

y = -x + 5



The lines meet at **(2, 3)**. Check that the point works in both equations:

First line: 3 = 2 + 1 ✓ **Second line:** 3 = -2 + 5 ✓

Because (2, 3) satisfies **both** equations, it is the solution to the system.

Try it together

1. If a point is on the blue line but not the orange line, is it a solution to the system? Why?

2. What happens when the spaghetti strands are parallel?

3. What happens when the two strands lie on top of one another?

4. Why is a graph useful when solving two equations?

Teacher note

Keep this lesson visual and conversational. Do not introduce elimination or substitution yet. First establish that a solution is a shared ordered pair, not simply where lines appear close together.